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Comprehensive Question Bank: Unit 5 — Data Science

Section A: Multiple-Choice Questions (MCQs)

Topic 5.1: Introduction to Data Science

1. What is Data Science best defined as?
   A) A field that only builds mobile apps
   B) The field where we use data to find patterns, answer questions, and help make better decisions
   C) A method of deleting unwanted files
   D) A type of computer hardware

2. In 1854, which physician used a dot map to trace a cholera outbreak to a single water pump?
   A) Edgar F. Codd
   B) Dr. John Snow
   C) Alan Turing
   D) Arthur Samuel

3. What did Dr. John Snow's cholera map help discover?
   A) A new disease
   B) That the outbreak was linked to a specific water pump
   C) A new type of database
   D) A new spreadsheet tool

4. According to the chapter, what is Dr. John Snow's map now considered the birth of?
   A) Modern databases
   B) Modern data visualization
   C) Modern spreadsheets
   D) Modern SQL

5. Why did Netflix feel confident that "House of Cards" would be a hit before filming it?
   A) They guessed based on personal opinion
   B) They analyzed years of user viewing data, including pauses, rewinds, and genres
   C) They flipped a coin
   D) They copied a competitor's show

6. What is the main risk of making decisions without using data?
   A) Decisions become too fast
   B) Decisions may be based on guesses or personal opinions, leading to mistakes
   C) Decisions always become perfect
   D) Data becomes automatically biased
7. Which of the following is an example of Data Science used in social media?
   A) Printing a newspaper
   B) Apps like Facebook and Instagram showing posts users are likely to enjoy
   C) Turning off a mobile phone
   D) Writing a letter by hand
8. Which of the following is an example of Data Science used in transport?
   A) A paper road map
   B) Google Maps using traffic data to suggest the fastest route
   C) A traffic light with no sensors
   D) A printed bus schedule
9. Which of the following is an example of Data Science used in healthcare?
   A) Manually filing paper prescriptions
   B) Hospitals predicting diseases and managing patient care using data
   C) Turning off hospital lights
   D) Printing medical textbooks
10. Which of the following is an example of Data Science used in education?
   A) Erasing a whiteboard
   B) Using student attendance and test result data to improve learning strategies
   C) Reading a printed textbook
   D) Turning off classroom projectors
11. E-commerce websites suggest products to users mainly based on:
   A) Random selection
   B) Data of the user's previous purchases
   C) Alphabetical order of product names
   D) The time of day only

Topic 5.2: Overview of the Data Science Life Cycle (Understanding the Problem)

1. What is the Data Science Life Cycle (DSLC)?
   A) A single-step process for solving problems
   B) A series of steps that help us solve a problem using data
   C) A type of database software
   D) A chart used only for visualization

2. Which of the following is the correct first step of the DSLC?
   A) Understanding the Problem
   B) Collecting Data
   C) Visualizing Results
   D) Cleaning and Validating Data

3. According to the chapter, what happens if we do not understand the problem properly before collecting data?
   A) The results automatically become more accurate
   B) We may collect the wrong data, resulting in useless or wrong results
   C) The data becomes automatically clean
   D) Nothing changes at all

4. Which of the following questions should be asked during the "Understanding the Problem" step?
   A) What chart should I use?
   B) What are we trying to find out?
   C) How do I clean the data?
   D) Which SQL query should I write?

5. Which of the following is a well-defined problem statement, according to the chapter's example?
   A) "Students are not happy."
   B) "Which school activities do students enjoy the most?"
   C) "School is boring."
   D) "Something is wrong somewhere."

6. When defining a problem, which of the following should be avoided?
   A) Clear and simple words
   B) One specific question
   C) Unclear or overly general terms
   D) A focused topic

7. What is the purpose of setting clear objectives in a data science project?
   A) To make the project more confusing
   B) To guide what data is needed, what questions to ask, and what results are expected
   C) To skip the data collection step
   D) To avoid using charts

8. According to Table 5.1, which objective matches the health-related problem "students not drinking enough water"?
   A) Suggest solutions to reduce late arrivals
   B) Collect data on how much water students drink daily
   C) Compare popularity and sales data
   D) Survey students on their favorite snacks

9. According to Table 5.1, which domain is associated with the problem "A school canteen wants to know which snacks students like the most"?
   A) School
   B) Health
   C) Business
   D) Transport

10. Which of the following best completes the sentence: "A good problem is one that can be _____ using data"?
   A) Ignored
   B) Answered by collecting and analyzing
   C) Deleted
   D) Encrypted

Topic 5.2: Collecting Data

1. What is described as "raw material" that must be gathered before cleaning, storing, or analyzing?
   A) A chart
   B) Data
   C) A database
   D) An objective

2. Which data collection method is described as the easiest and most common?
   A) Web Scraping
   B) Surveys
   C) IoT Sensors
   D) APIs

3. What does the acronym API stand for?
   A) Automated Program Interaction
   B) Application Programming Interface
   C) Applied Programming Index
   D) Automatic Public Information
4. Which data collection method allows software to automatically pull data from websites or apps?
   A) Surveys
   B) APIs
   C) Web Scraping only
   D) Manual paper forms
5. What does IoT stand for?
   A) Internet of Technology
   B) Internet of Things
   C) Input or Output Tool
   D) Integrated Online Tracker
6. Which of the following is an example of IoT data collection mentioned in the chapter?
   A) A paper survey form
   B) Smart water meters placed near fountains to study water-drinking habits
   C) A printed report
   D) A whiteboard drawing
7. What is web scraping?
   A) Manually copying data by hand
   B) The process of programmatically collecting public data from websites
   C) Deleting data from a website
   D) Encrypting website data
8. According to the chapter, web scraping should only be done:
   A) On any website without restriction
   B) With teacher guidance, and only on websites that allow it
   C) Without checking website permissions
   D) Only at night
9. When choosing a data source, which of the following questions should you NOT primarily rely on?
   A) Is this data from a trusted place?
   B) Is it recent or outdated?
   C) Is it the most colorful data available?
   D) Does it match the problem I want to solve?

10. If you want to know students' favorite snacks in 2025, why should you avoid using a 2020 survey?
    A) Because 2020 surveys are always fake
    B) Because it is too outdated and may not reflect current preferences
    C) Because old surveys are illegal
    D) Because paper surveys cannot be trusted
11. Data quality means the data is complete, correct, and:
    A) Colorful
    B) Relevant (matches the problem)
    C) Expensive
    D) Encrypted
12. What does "bias" mean in the context of data collection?
    A) Data that is well organized
    B) Data that only shows part of the truth and misses the full picture
    C) Data that is very large in size
    D) Data collected using APIs only
13. If a survey about favorite sports only asks boys, what is the likely result?
    A) A perfectly balanced result
    B) A biased result that misses girls' opinions
    C) A result with no data at all
    D) An automatically corrected result
14. Which of the following helps avoid bias in data collection?
    A) Asking only your close friends
    B) Asking a diverse group of people
    C) Using leading questions
    D) Ignoring strange patterns in answers

Topic 5.2: Cleaning and Validating Data

1. What is "raw data"?
   A) Data that has already been cleaned
   B) The original, uncleaned information collected from people, sensors, websites, or apps
   C) Data stored only in a database
   D) A type of chart

2. In Table 5.2, "Footbal" is an example of which type of data error?
   A) Missing data
   B) A spelling mistake
   C) A duplicate entry
   D) An outlier
3. In Table 5.2, a blank cell where a sport name should be is an example of:
   A) A spelling mistake
   B) Missing data
   C) An outlier
   D) A duplicate entry
4. "Cricket", "cricket", and "CRICKET" appearing as three different entries is an example of:
   A) Missing data
   B) Inconsistent formatting
   C) An outlier
   D) A duplicate entry
5. A frequency value written as "1000 times/week" is an example of:
   A) Missing data
   B) A duplicate entry
   C) An outlier (a strange or unrealistic value)
   D) Inconsistent formatting
6. What does "data validation" mean?
   A) Deleting all data permanently
   B) Checking if data follows correct rules, has the right value types, and is not fake or silly
   C) Adding colors to a chart
   D) Translating data into another language
7. Which of the following is a recommended validation technique for surveys?
   A) Use only blank text boxes
   B) Use multiple choice options instead of blank text boxes
   C) Make all questions optional
   D) Avoid using dropdowns
8. According to the chapter, what percentage of their time do professional data scientists often spend cleaning data?
   A) About 10%
   B) About 80%
   C) About 30%
   D) About 50%

9. Which of the following is the correct fix for the spelling error "Footbal"?
   A) "Foutball"
   B) "Football"
   C) "Footbol"
   D) Leave it unchanged
10. What is the main risk of analyzing data without cleaning it first, according to the chapter's example?
   A) The chart will automatically look better
   B) Misspelled duplicates like "cricket" and "Cricket" may be counted separately, giving a wrong picture
   C) The data will delete itself
   D) SQL queries will stop working
11. Which of the following best summarizes the chapter's conclusion about data cleaning?
   A) Dirty data leads to smart decisions
   B) Clean data leads to smart decisions; dirty data leads to incorrect results
   C) Cleaning data is unnecessary
   D) Only databases need cleaning, not spreadsheets

Topic 5.2: Analyzing and Interpreting Data

1. What does it mean to "analyze data"?
   A) To delete unnecessary rows
   B) To carefully examine data to summarize it, compare groups, and find patterns
   C) To only store data in a database
   D) To translate data into another format
2. Which of the following is a simple measure used in data analysis, according to the chapter?
   A) Encryption
   B) Frequency (how often)
   C) Compression
   D) Duplication
3. According to Table 5.4, which sport was most preferred by the surveyed students?
   A) Cricket
   B) Football
   C) Hockey
   D) Badminton

4. What is the main purpose of the "Interpreting Data" step?
   A) To collect more raw data
   B) To find meaning from data by identifying trends and patterns
   C) To design a new survey
   D) To install spreadsheet software
5. In the "Screen Time" analogy from the chapter, what does the raw number "4 hours on TikTok" represent?
   A) Data
   B) Interpretation
   C) A database record
   D) A validation rule
6. In the "Screen Time" analogy, realizing you should put your phone away before bed represents:
   A) Raw data
   B) Interpretation
   C) Data collection
   D) Data validation
7. If most late-arriving students are found to live far from school, what solution might we recommend?
   A) Cancel school
   B) Provide transport support
   C) Delete the attendance data
   D) Stop analyzing data

Topic 5.2: Visualizing and Communicating Results

1. Why do we use data visualizations, according to the chapter?
   A) Because numbers are always more useful than pictures
   B) Because people usually understand pictures better than numbers
   C) Because visualizations are required by law
   D) Because they remove the need for data collection
2. Which of the following is NOT listed as a benefit of data visualization?
   A) Makes data easier to understand
   B) Shows patterns at a glance
   C) Automatically fixes all data errors
   D) Helps share ideas clearly in presentations

3. According to Table 5.7, which chart type is best for comparing groups?
   A) Pie Chart
   B) Bar Chart
   C) Line Chart
   D) Scatter Chart
4. According to Table 5.7, which chart type is best for showing parts of a whole?
   A) Bar Chart
   B) Pie Chart
   C) Line Chart
   D) Dot Chart
5. According to Table 5.7, which chart type is best for showing changes over time?
   A) Bar Chart
   B) Pie Chart
   C) Line Chart
   D) Table Chart
6. Which of the following is a properly written result summary, according to the chapter's style?
   A) "Data numbers exist."
   B) "Most students (10) like cricket, followed by football (5). Very few students like badminton (1)."
   C) "Cricket is a sport."
   D) "Charts are pictures."
7. What is the main risk illustrated by a bar chart whose Y-axis starts at 50 instead of 0?
   A) The chart becomes impossible to draw
   B) Small differences between groups can appear misleadingly large
   C) The chart automatically becomes a pie chart
   D) The data becomes encrypted

Topic 5.3: Tools for Visualization

1. Which two tools are specifically mentioned as simple yet powerful tools for creating charts?
   A) Python and Java
   B) Google Sheets and Microsoft Excel
   C) MySQL and SQLite
   D) Photoshop and Illustrator

2. Which of the following is NOT a reason given for using spreadsheets to create charts?
   A) Easy to use
   B) No installation needed (for Google Sheets)
   C) Requires advanced coding knowledge
   D) Good for small to medium-sized projects
3. In Google Sheets, which menu do you click to insert a chart?
   A) File
   B) Insert
   C) Format
   D) Data
4. After selecting "Insert" and "Chart" in Google Sheets, what should you do next to change the chart type?
   A) Close the program
   B) Select the desired chart type (e.g., bar chart) from the chart editor panel
   C) Delete the data
   D) Save the file as a PDF only
5. What should always be added to a chart to make it understandable to others?
   A) Random colors only
   B) A clear title and labeled axes
   C) Background music
   D) Extra unrelated data
6. What is a "dashboard," as mentioned in the chapter?
   A) A single number with no visuals
   B) A display that shows multiple visuals in one place to tell a clear data story
   C) A type of spreadsheet formula
   D) A database table

Topic 5.4: Databases and Storing Data

1. Who is credited with inventing the Relational Database in 1970?
   A) Alan Turing
   B) Edgar F. Codd
   C) Dr. John Snow
   D) Arthur Samuel

2. Where did Edgar F. Codd work when he proposed the Relational Database?
   A) Google
   B) IBM
   C) Microsoft
   D) Netflix
3. What problem was Edgar F. Codd trying to solve with the Relational Database?
   A) Slow internet speeds
   B) Messy, disorganized digital filing cabinets that made data hard to find or connect
   C) A lack of chart types
   D) Poor survey response rates
4. According to Table 5.8, spreadsheets are best suited for:
   A) Small datasets
   B) Large, multi-user data systems
   C) Only image files
   D) Only video files
5. According to Table 5.8, databases are best suited for:
   A) Small personal notes only
   B) Structured systems that store and manage large data in tables
   C) Only paper-based records
   D) Only single-user situations
6. Which of the following is an example tool for a database, according to Table 5.8?
   A) Excel
   B) MySQL
   C) Google Sheets
   D) PowerPoint
7. What is a database, according to the chapter's definition?
   A) A single column of numbers
   B) An organized collection of data, stored in the form of tables
   C) A chart showing trends
   D) A type of survey
8. According to Table 5.9, what is a "Table" in database terminology?
   A) A group of related data
   B) A single row of data
   C) A single piece of information
   D) A type of chart

9. According to Table 5.9, what is a "Record" in database terminology?
   A) A group of related data
   B) A single entry (row) in the table
   C) A specific column heading
   D) A type of SQL query
10. According to Table 5.9, what is a "Field" in database terminology?
   A) A full table
   B) A full row
   C) A specific piece of information (a column)
   D) A chart type
11. What does DBMS stand for?
   A) Data Backup Management Software
   B) Database Management System
   C) Digital Base Management Server
   D) Data Bank Monitoring System
12. Which of the following is an example of a DBMS mentioned in the chapter?
   A) Google Sheets
   B) MySQL
   C) Google Data Studio
   D) Google Forms
13. What does SQL stand for?
   A) Simple Question Language
   B) Structured Query Language
   C) System Quality Log
   D) Sequential Query List
14. Which SQL command is used to retrieve data from a table?
   A) SELECT
   B) INSERT
   C) DELETE
   D) UPDATE
15. Which SQL command is used to add new data to a table?
   A) SELECT
   B) INSERT
   C) REMOVE
   D) FILTER

16. What does the query `SELECT * FROM Sports;` do?
    A) Deletes all data from the Sports table
    B) Shows all the data from the table named Sports
    C) Adds a new entry to the Sports table
    D) Renames the Sports table

17. What does the query `SELECT Name FROM Sports WHERE Favorite_Sport = 'Cricket';` do?
    A) Deletes students who like Cricket
    B) Shows names of students in the Sports table who like Cricket
    C) Adds a new student named Cricket
    D) Renames the Favorite_Sport column

18. What does the query `INSERT INTO Sports (Name, Favorite_Sport) VALUES ('Ali', 'Football');` do?
    A) Deletes Ali's record
    B) Displays all records in the Sports table
    C) Adds a new entry for Ali with Favorite_Sport as Football
    D) Updates Ali's existing record

19. Why might storing hospital patient records in an ordinary spreadsheet be risky, according to the chapter's reasoning?
    A) Spreadsheets are always more secure than databases
    B) Spreadsheets lack the structured, secure, multi-user management that databases provide
    C) Spreadsheets cannot store numbers
    D) Spreadsheets are illegal to use

Section B: Short Answer Questions (SQs)

Topic 5.1: Introduction to Data Science

1. Define Data Science in your own words.
2. Explain briefly how Dr. John Snow's 1854 cholera map relates to modern data visualization.
3. Explain briefly how Netflix used data to predict the success of "House of Cards."
4. State two real-life applications of Data Science mentioned in the chapter.
5. Explain briefly why decisions based on data are generally better than decisions based on guesses.

Topic 5.2: Understanding the Problem

1. Define what it means to "understand the problem" in the Data Science Life Cycle.
2. List the three key questions (what, why, who) asked during the problem understanding step.
3. Differentiate between a poorly defined problem and a clearly defined problem, using an example.
4. Explain briefly why setting clear objectives is important in a data science project.
5. Identify the domain (School, Health, or Business) for the following problem: "A canteen wants to know which snacks students like the most," and state one matching objective.

Topic 5.2: Collecting Data

1. Define "data collection" and explain why it is necessary after understanding a problem.
2. List and briefly describe two methods of data collection mentioned in the chapter.
3. Differentiate between an API and Web Scraping as data collection methods.
4. Explain briefly how IoT devices can be used to collect data, with one example.
5. Identify what questions you should ask when choosing a reliable data source.
6. Define "bias" in data collection, and explain briefly how it can affect results.
7. State two ways to avoid bias when collecting data.

Topic 5.2: Cleaning and Validating Data

1. Define "raw data" in your own words.
2. Differentiate between a "missing data" error and a "duplicate entry" error, using examples.
3. Explain briefly what "data validation" means.
4. List two techniques for validating data collected through surveys.
5. Map out, in order, at least three steps involved in cleaning messy data.
6. Explain briefly why treating "cricket" and "Cricket" as two different values could lead to incorrect analysis results.

Topic 5.2: Analyzing and Interpreting Data

1. Define what it means to "analyze data."
2. State two simple tools that can be used for basic data analysis.
3. Differentiate between "analyzing data" and "interpreting data," using the Screen Time example.
4. Explain briefly how a data scientist might interpret the finding that most late-arriving students live far from school.
5. Predict the likely conclusion if a survey shows Cricket = 10, Football = 5, Hockey = 4, Badminton = 1.

Topic 5.2: Visualizing and Communicating Results

1. Define "data visualization" and explain why it is used.
2. Differentiate between a Bar Chart, a Pie Chart, and a Line Chart in terms of their best use case.
3. Explain briefly how a bar chart with a Y-axis starting at 50 instead of 0 could mislead a reader.
4. State the two components that should always be included when communicating results (visuals and what else?).

Topic 5.3: Tools for Visualization

1. Name two spreadsheet tools mentioned in the chapter for creating charts.
2. List three reasons why spreadsheets are considered good tools for beginners.
3. Map out, in order, the steps needed to create a bar chart in Google Sheets.
4. Define what a "dashboard" is, according to the chapter.

Topic 5.4: Databases and Storing Data

1. Define what a database is, in your own words.
2. Differentiate between a spreadsheet and a database, using at least two points of comparison.
3. Define the terms "Table," "Record," and "Field" as used in databases.
4. Explain briefly who Edgar F. Codd was and what problem his invention solved.
5. Define DBMS and give one example.
6. Define SQL and explain briefly what it is used for.
7. Identify what the SQL command `SELECT * FROM Sports;` would do.
8. Identify what the SQL command `INSERT INTO Sports (Name, Favorite_Sport) VALUES ('Ali', 'Football');` would do.

Section C: Long / Extensive Questions (LQs)

Topic 5.1: Introduction to Data Science

1. Discuss in detail the concept and importance of Data Science in everyday life.
   a) Define Data Science and explain how it turns raw data into useful information.
   b) Describe, using the story of Dr. John Snow, how data visualization can solve real-world problems even without modern technology.
   c) Explain, using the Netflix "House of Cards" example, how data can be used to predict

outcomes before they happen.
d) Discuss at least three real-life applications of Data Science across different domains (e.g., social media, healthcare, transport, education).

Topic 5.2: Overview of the Data Science Life Cycle

1. Elaborate on the complete Data Science Life Cycle (DSLC) from problem understanding to visualization.
   a) List and briefly explain all six steps of the DSLC in the correct order.
   b) Using the "empty school library on Fridays" example, construct a full step-by-step trace of how each of the six DSLC steps would be applied to solve this real problem.
   c) Explain why an additional seventh step, "Storing and Organizing Data," is sometimes added in real-world, large-scale projects.

2. Analyze the process of collecting reliable and unbiased data.
   a) Describe the four main methods of data collection discussed in the chapter (Surveys, APIs, IoT/Sensors, Web Scraping), including one example of each.
   b) Discuss the criteria that should be used to judge whether a data source is trustworthy and appropriate for a given problem.
   c) Evaluate the concept of bias in data collection, and design two strategies a student could use to minimize bias in a school-based survey project.

3. Construct a detailed explanation of data cleaning and validation, supported by a worked example.
   a) Explain why raw data is often messy, and justify why data cleaning is considered a normal and essential part of data science work.
   b) Using a table similar to Table 5.2, identify at least four different types of data errors and explain how each would be corrected.
   c) Design a complete data validation checklist that a student could use before analyzing any survey dataset.

4. Discuss in detail the process of analyzing, interpreting, and visualizing data to reach a final decision.
   a) Differentiate between "analyzing" and "interpreting" data, explaining why both steps are necessary and cannot be skipped.
   b) Given a sample frequency table of favorite sports (Cricket = 10, Football = 5, Hockey = 4, Badminton = 1), construct a full written interpretation and a bar chart description of the findings.
   c) Evaluate why some data visualizations can be misleading (e.g., a Y-axis that does not start at zero), and discuss best practices to avoid creating misleading charts.

Topic 5.3: Tools for Visualization

1. Discuss in detail how spreadsheet tools like Google Sheets and Excel are used to visualize data.
   a) Explain why spreadsheets are considered beginner-friendly tools for data visualization.
   b) Construct a complete, numbered, step-by-step guide for turning a sample frequency table into a bar chart in Google Sheets, from entering data to customizing the final chart.
   c) Justify why properly labeling a chart's title and axes is just as important as the data itself, using a real-world scenario where an unlabeled chart could cause confusion.

Topic 5.4: Databases and Storing Data

1. Discuss in detail the role of databases in storing and managing structured data.
   a) Explain the historical origin of the Relational Database through Edgar F. Codd's work at IBM in 1970, and discuss the problem it was designed to solve.
   b) Construct a detailed comparison table between Spreadsheets and Databases, covering structure, scale, and appropriate use cases.
   c) Define and differentiate between the terms Table, Record, and Field, using a concrete student-database example.

2. Design a complete database system and demonstrate its use with SQL.
   a) Given a messy paragraph describing three students (name, age, and grade), construct a clean, structured database table with appropriate columns.
   b) Write and explain three different SQL queries (using SELECT and INSERT) that could be used to retrieve or add data to your constructed table.
   c) Evaluate the risks of storing sensitive information, such as hospital patient records, in an ordinary spreadsheet instead of a secure, structured database, and justify why a database would be the safer choice.